

# HESOD: HIGH-RESOLUTION EFFICIENT SMALL OBJECT DETECTION VIA SEMANTIC-SPECTRAL DUAL SELECTION

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## ABSTRACT

Tiny-object detection in high-resolution aerial and maritime imagery must contend with vast, redundant backgrounds and extreme scale variation. While spatial selective computation routes foreground regions to cut latency, existing single-evidence selectors suffer severe semantic collapse on micro targets, causing irrecoverable pruning. We propose **HESOD** (High-Resolution Efficient Small Object Detection), a dual-evidence framework unifying semantic-spectral routing, coverage-constrained optimization, and lightweight detection. HESOD couples semantic objectness with constant-width spectral saliency under a monotonic evidence-preserving max-fusion rule to rescue degraded targets. An object-level soft-coverage loss optimizes multi-cell instance coverage across ground-truth objects for recall-safe routing. To recover downstream efficiency, an Inverse-Residual Shared-Parameter Partial-Convolution (ISPP) head compresses parameters by  $>27\%$  with only fractional FLOPs overhead. Experiments show that HESOD improves mAP@.5 by 0.9 pp and 2.3 pp (boosting total instance recall by 3.66 pp and 3.33 pp) on UAVDT and SeaPerson, respectively, at up to 106 FPS, making it well suited to resource-constrained deployment.

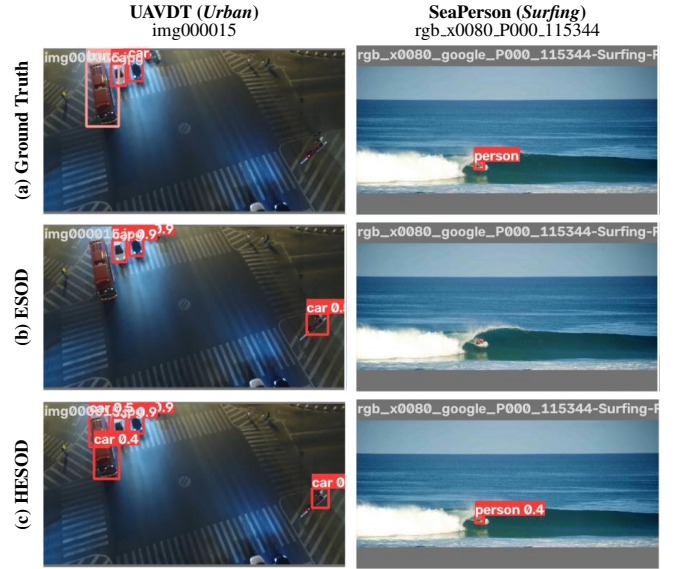
**Index Terms**— Small object detection, selective computation, spectral saliency, coverage optimization, UAV

## 1. INTRODUCTION

High-resolution aerial and maritime vision is fundamental to intelligent surveillance, environmental monitoring, and search-and-rescue. In operational scenarios across aerial and maritime domains [1–3], targets of interest typically occupy only a handful of pixels ( $<16 \times 16$  px) amid massive uniform backgrounds ( $>70\%$ – $80\%$ ) [4]. Evaluating standard dense deep detectors across multi-megapixel grids ( $1280 \times 1280$  to  $2048 \times 2048$ ) incurs prohibitive arithmetic latency on resource-constrained platforms.

To eliminate redundant computation, *spatial selective computation* routes feature processing to prospective foreground regions [5, 6]. Notably, ESOD [7] predicts an early spatial priority heatmap after a shallow stem, slicing features into candidate patches (AdaSlicer) and bypassing background across downstream stages.

However, spatial selection operates under a severe *asymmetric error cost*: false-positive background patches merely incur minor compute overhead, whereas discarding a true tiny target produces an *irrecoverable false negative*. Existing detectors rely on single-semantic objectness with independent per-cell BCE. Under extreme



**Fig. 1.** Visual comparison on UAVDT and SeaPerson. Baseline ESOD drops micro targets (b), while HESOD restores full instance recall (c).

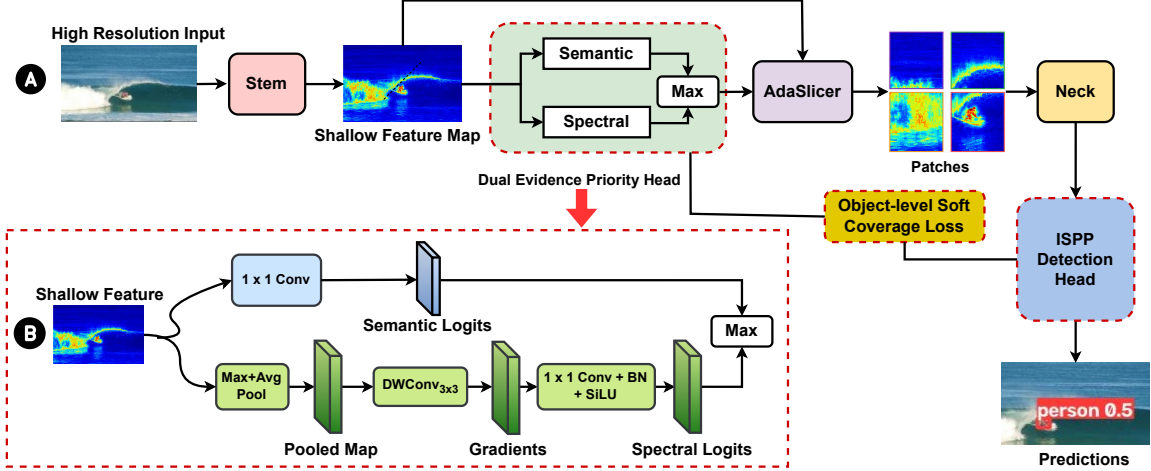
downsampling, micro targets suffer severe semantic collapse, causing faint instances to be prematurely pruned (Fig. 1(b)). To overcome this recall dilemma, we observe that tiny targets retain high-frequency spatial gradients and spectral saliency even when deep semantics fade [8–10]. We then propose High-Resolution Efficient Small Object Detection (**HESOD**) to establish recall-safe, evidence-preserving sparse routing. By coupling dual-evidence routing with an object-level soft-coverage loss, our selector reliably rescues these dropped targets (Fig. 1(c)). Our contributions include:

1. **Dual-Evidence Spatial Routing:** We uncover single-semantic routers’ vulnerability to feature collapse, coupling semantic objectness with constant-width gradient filtering for scale-robust structural priors independent of backbone capacity.
2. **Evidence Preservation and Soft Coverage:** We formulate a monotonic max-fusion rule preventing multi-cue evidence dilution, paired with an object-level noisy-OR soft-coverage loss directly optimizing joint instance recall.
3. **Compute Compensation and Empirical Gains:** We integrate an ISPP head that cuts detector parameters by  $>27\%$  while largely retaining dual-evidence routing’s gains; the resulting HESOD improves mAP@.5 by 0.9/2.3 pp and total instance recall by 3.66/3.33 pp on UAVDT/SeaPerson.

\* Source code at <https://github.com/deepsota-ai/HESOD>.

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**Fig. 2.** Overview of HESOD: (a) End-to-end selective pipeline; (b) Dual-Evidence Priority Head. Red dashed boxes highlight our proposed contributions over ESOD.

## 2. RELATED WORK

### 2.1. Small Object Detection in High-Resolution Vision

Detecting tiny targets in high-resolution aerial and maritime imagery is severely challenged by extreme scale disparity, low contrast, and massive background areas ( $>70\%$ – $80\%$ ) [1–3, 11]. Classical approaches utilize density-guided cropping (ClusDet [12], DMNet [13]) or scale-adaptive label assignment (NWD [14], RFLA [15]). Recently, ResBiDet [16] introduced a coarse-to-fine framework using global heatmap guidance for region cropping. However, it relies on multi-scale image-level slicing and boundary stitching. In contrast, HESOD performs early feature-level selective routing directly within the detector trunk and explicitly optimizes instance coverage under irreversible pruning.

### 2.2. Selective and Dynamic Computation

Dynamic computation accelerates inference by allocating compute strictly to prospective foreground regions (e.g., AutoFocus [4], QueryDet [5], CEASC [6], and sparse location activation [17]). Recently, coarse-to-fine detectors such as ESOD [7] and OSDT [18] predict early spatial priority heatmaps to extract candidate patches. However, existing selectors rely exclusively on a single semantic score supervised by independent per-cell BCE. Under extreme scale disparity, micro targets suffer from semantic feature collapse after downsampling; meanwhile, per-cell BCE lacks joint instance-level coverage constraints, causing fatal false negatives under an asymmetric error cost. HESOD resolves this by combining semantic and spectral dual cues with object-level soft-coverage optimization.

### 2.3. Spectral Saliency and Frequency Methods

High-frequency edge gradients persist for micro targets even as photometric semantics fade; recent methods exploit this via dense frequency transforms [8], frequency-gradient networks [10], dual-domain alignment [9], or multi-scale compensation [19], but all apply spectral cues *downstream*, incurring substantial latency. In sharp contrast, HESOD repurposes high-frequency gradient saliency *upstream* as an ultra-lightweight routing cue: channel pooling compresses stem features to 2 channels, enabling directional Laplacian and Sobel gradient filtering at constant width, at negligible overhead.

## 3. PROPOSED METHOD

The architecture of HESOD is shown in Fig. 2. From input  $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ , a shallow stem extracts dense features  $\mathbf{F} \in \mathbb{R}^{H_s \times W_s \times C}$ . The *Dual-Evidence Spatial Selector* then coordinates: (i) a priority head fusing semantic objectness with channel-pooled spectral saliency (Fig. 2(b)) into spatial priority map  $\mathbf{S} \in [0, 1]^{H_s \times W_s}$ ; and (ii) an AdaSlicer module (unchanged from ESOD [7]) clustering candidate foreground cells into bounding patches  $\mathcal{P}$ . Downstream backbone stages and lightweight decoupled heads evaluate only retained patches  $\mathcal{P}$ , trained under the soft-coverage supervision (Sec. 3.2) via the two-stage protocol (Sec. 3.3).

### 3.1. Dual-Evidence Priority Estimation

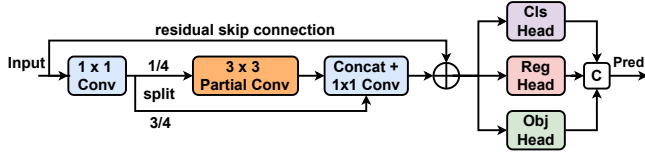
To rescue micro targets with degraded semantic activations, HESOD couples high-level semantic confidence with low-level structural gradient saliency at negligible compute overhead (Fig. 2(b)).

Let  $\mathbf{f}_i \in \mathbb{R}^C$  denote the feature vector at spatial location  $i$  in  $\mathbf{F}$ . The semantic branch predicts class-specific logits via a  $1 \times 1$  convolution:  $\mathbf{z}_i^{\text{sem}} = \text{Conv}_{1 \times 1}(\mathbf{f}_i) \in \mathbb{R}^{N_c}$ . In parallel, the spectral branch extracts high-frequency structural gradients by first compressing channels via max- and mean-pooling:

$$\mathbf{F}_{\text{pooled}} = \left[ \max_c \mathbf{F}_{:, :, c} \parallel \frac{1}{C} \sum_{c=1}^C \mathbf{F}_{:, :, c} \right] \in \mathbb{R}^{H_s \times W_s \times 2}, \quad (1)$$

where  $\mathbf{F}_{:, :, c} \in \mathbb{R}^{H_s \times W_s}$  is the  $c$ -th channel of  $\mathbf{F}$  and  $\parallel$  denotes channel-wise concatenation. Compressing  $\mathbf{F}$  to 2 channels retains peak edge activation and background energy, after which a learnable depthwise  $3 \times 3$  convolution bank—initialized with discrete Laplacian and directional Sobel operators—filters  $\mathbf{F}_{\text{pooled}}$  into 6 structural channels at  $\mathcal{O}(H_s W_s)$  cost independent of width  $C$ . In the 2D spatial frequency domain, the discrete Laplacian is approximately isotropic in the low-frequency limit, with  $|H(\omega)| \propto \|\omega\|^2$ , naturally amplifying edge transitions. Because micro targets ( $<16^2$  px) manifest as localized high-frequency spatial impulses whose gradient energy resists downsampling [10], this branch supplies a constant-width prior independent of deep semantic capacity. A  $1 \times 1$  projection maps features to  $N_c$ , yielding spectral logits  $\mathbf{z}_i^{\text{spec}} \in \mathbb{R}^{N_c}$ .

To prevent confidence dilution on faint targets, evidence fusion enforces *evidence preservation*. Linear combinations risk destructive interference: an unconfident or negative activation in one branch



**Fig. 3.** Inverse-Residual Shared-Parameter Partial-Convolution (ISPP) Head (⊕ denotes channel concatenation).

can drag down a confident signal in the other below the routing threshold. Instead, we formulate a parameter-free elementwise maximum:

$$z_{i,c} = \max(z_{i,c}^{\text{sem}}, z_{i,c}^{\text{spec}}), \quad s_i = \max_{c=1}^{N_c} \sigma(z_{i,c}), \quad (2)$$

This operation functions as an evidence-preserving OR gate: since  $z_{i,c} \geq z_{i,c}^{\text{sem}}$  and  $z_{i,c} \geq z_{i,c}^{\text{spec}}$ , neither branch can ever suppress genuine candidate cues. Cells with priority  $s_i > \tau$  are clustered into bounding patches  $\mathcal{P}$  for downstream detection.

### 3.2. Object-Level Soft-Coverage Loss

Standard pixel-wise binary cross-entropy (BCE) evaluates grid cells independently, failing to guarantee instance-level coverage: a selector may over-allocate cells to large objects while dropping micro objects whose receptive fields contain only weak responses. To prevent premature false negatives, we formulate a differentiable object-level soft-coverage loss.

Let  $\mathcal{N}(j)$  denote the set of selector cells overlapping ground-truth box  $j \in \{1, \dots, N_{\text{gt}}\}$ . We formulate a noisy-OR differentiable coverage surrogate and its weighted instance loss:

$$p_j^{\text{cover}} = 1 - \prod_{i \in \mathcal{N}(j)} (1 - s_i), \quad \ell_j = -w_j \log(p_j^{\text{cover}} + \epsilon), \quad (3)$$

where  $w_j = \text{clip}(4/a_j, 1, 5)$  is a scale-adaptive weight based on bounding box area  $a_j$  in cells, penalizing misses on micro targets ( $a_j < 4$ ). For any cell  $i \in \mathcal{N}(j)$ , the gradient of  $\ell_j$  with respect to cell score  $s_i$  is:

$$\frac{\partial \ell_j}{\partial s_i} = -\frac{w_j}{p_j^{\text{cover}} + \epsilon} \prod_{k \in \mathcal{N}(j) \setminus \{i\}} (1 - s_k). \quad (4)$$

When neighboring candidate cells have low scores ( $s_k \rightarrow 0$ ), the product term approaches 1, generating a strong gradient to boost cell  $i$  and encouraging at least one cell to activate per instance. The overall soft-coverage loss averages over instances:  $\mathcal{L}_{\text{cover}} = \frac{1}{N_{\text{gt}}} \sum_{j=1}^{N_{\text{gt}}} \ell_j$ . The multi-task training objective is  $\mathcal{L} = \mathcal{L}_{\text{det}} + \lambda_{\text{sel}}(\mathcal{L}_{\text{BCE}} + \lambda_{\text{cov}} \mathcal{L}_{\text{cover}})$ , where  $\mathcal{L}_{\text{BCE}}$  is the per-cell selector BCE over scores  $s_i$ , with  $\lambda_{\text{sel}} = 0.2$  and  $\lambda_{\text{cov}} = 0.5$ .

### 3.3. Lightweight Decoupled Head and Training Protocol

While dual-evidence routing substantially improves patch coverage by retaining faint foreground regions, evaluating more candidate patches increases downstream detector computation. To recover this overhead without sacrificing selector gains, HESOD incorporates an ISPP decoupled head [20] as a dedicated compute compensator.

As shown in Fig. 3, the ISPP head expands neck feature channels of width  $C_1$  to  $2C_1$ , partitioning them into an active branch (0.25 ratio) processed by a  $3 \times 3$  Partial Convolution (PConv) and an identity passthrough branch (0.75 ratio). After linear projection and residual addition, decoupled  $1 \times 1$  heads predict classification, box regression, and objectness independently, lowering arithmetic complexity while eliminating inter-task gradient interference.

**Two-Stage Training Protocol.** To ensure stable routing behavior, HESOD employs a two-stage training scheme: (i) *Stage 1*

**Table 1.** Benchmark results on UAVDT and SeaPerson.

| Method                                   | mAP@.5         | Params (M)    | GFLOPs         | Total Recall    | BPR            | FPS          |
|------------------------------------------|----------------|---------------|----------------|-----------------|----------------|--------------|
| <b>UAVDT Benchmark</b> (1280 × 1280)     |                |               |                |                 |                |              |
| Faster R-CNN [21]                        | 0.336          | 43.3          | 772.5          | 79.68%          | —              | 18.8         |
| RetinaNet [22]                           | 0.296          | 36.4          | 360.9          | 83.44%          | —              | 19.8         |
| QueryDet <sup>†</sup> [5]                | 0.257          | 39.3          | — <sup>†</sup> | 41.89%          | —              | 21.1         |
| ESOD [7]                                 | <u>0.385</u>   | <u>35.9</u>   | <b>68.2</b>    | <u>85.17%</u>   | <u>0.884</u>   | <b>117.8</b> |
| <b>HESOD (Ours)</b>                      | <b>0.394</b>   | <b>26.0</b>   | <u>74.9</u>    | <b>88.83%</b>   | <b>0.940</b>   | <u>106.1</u> |
| Δ vs. ESOD                               | <b>+0.9 pp</b> | <b>-27.5%</b> | +9.8%          | <b>+3.66 pp</b> | <b>+5.6 pp</b> | -9.9%        |
| <b>SeaPerson Benchmark</b> (2048 × 2048) |                |               |                |                 |                |              |
| Faster R-CNN [21]                        | 0.551          | 43.3          | 1546.8         | 67.27%          | —              | 23.1         |
| RetinaNet [22]                           | 0.473          | 36.4          | 942.7          | 65.98%          | —              | 28.0         |
| QueryDet <sup>†</sup> [5]                | 0.627          | 39.3          | — <sup>†</sup> | 61.56%          | —              | 14.0         |
| ESOD [7]                                 | <u>0.750</u>   | <u>35.8</u>   | <b>202.4</b>   | <u>84.42%</u>   | <u>0.947</u>   | <b>85.7</b>  |
| <b>HESOD (Ours)</b>                      | <b>0.773</b>   | <b>25.9</b>   | <u>208.5</u>   | <b>87.75%</b>   | <b>0.991</b>   | <u>80.6</u>  |
| Δ vs. ESOD                               | <b>+2.3 pp</b> | <b>-27.6%</b> | +3.0%          | <b>+3.33 pp</b> | <b>+4.4 pp</b> | -6.0%        |

BPR: Best Possible Recall for objects' bounding boxes (BPR<sup>box</sup>), following ESOD [7]. **Bold**: best; underline: second-best (selective models).

<sup>†</sup>QueryDet uses CSQ (sparse FLOPs unamenable to static profiling; dense query counterparts: 973.5 / 2539.0 GFLOPs at 12.5 / 10.6 FPS on UAVDT / SeaPerson).

(*Selector Convergence*): Train the Dual-Max selector with a coupled head under  $\mathcal{L}_{\text{cover}}$ , establishing a well-calibrated priority heatmap; and (ii) *Stage 2 (Head Fine-Tuning)*: Freeze the upstream trunk (backbone, dual branches, router) and fine-tune the lightweight ISPP neck and decoupled heads on the frozen routing distribution. Relative to the uncompressed Dual-Max selector, this lightweight head slashes GFLOPs by 16.9% and parameters by 27.5% (on UAVDT) while fully preserving peak selector coverage (0.940), maintaining competitive real-time FPS with only fractional FLOPs overhead vs. ESOD.

## 4. EXPERIMENTS

### 4.1. Experimental Setup

We evaluate HESOD on two demanding high-resolution small-object benchmarks: **UAVDT** [2] (1280 × 1280 px, urban vehicle surveillance across 3 classes, 19.9% extremely tiny targets <16<sup>2</sup> px) and **SeaPerson** [3, 11] (2048 × 2048 px, maritime search-and-rescue with >85% targets <32 px). Both benchmarks use full-resolution images without artificial cropping.

Selective detectors (ESOD family and our ablations) employ YOLOv5m backbones in PyTorch 2.8 trained for 50 epochs under selective protocols [7]. Dense and query baselines (Faster R-CNN, RetinaNet, QueryDet) retain canonical ResNet-50-FPN backbones, evaluated on identical test splits and resolutions. All models are benchmarked on an NVIDIA RTX 5090 GPU. Metrics include COCO mAP@.5 and mAP@.5 : .95, scale-stratified recall across four bins (Very Tiny <16<sup>2</sup>, Tiny 16<sup>2</sup>–32<sup>2</sup>, Small 32<sup>2</sup>–96<sup>2</sup>, Med/Large >96<sup>2</sup> px), and, following ESOD [7], Best Possible Recall for objects' bounding boxes (BPR<sup>box</sup>), measuring the fraction of ground-truth targets covered (>50% overlap) by selected patches.

### 4.2. Main Benchmark Comparisons

As shown in Table 1, traditional dense detectors suffer from severe efficiency bottlenecks on high-resolution images: Faster R-CNN only achieves 18.8/23.1 FPS on UAVDT/SeaPerson with GFLOPs as high as 772.5/1546.8; QueryDet [5] is limited by its dynamic query mechanism, with total recall dropping to only 41.89% on UAVDT. HESOD outperforms all baselines: on UAVDT, mAP@.5 is 5.8,

**Table 2.** Module ablation, efficiency, and scale recall breakdown on UAVDT and SeaPerson.

| Configuration                                     | Detection Accuracy and Computational Efficiency |                |              |               |              |              | Physical-Scale Recall Breakdown |                                 |                                  |                         |                 |
|---------------------------------------------------|-------------------------------------------------|----------------|--------------|---------------|--------------|--------------|---------------------------------|---------------------------------|----------------------------------|-------------------------|-----------------|
|                                                   | mAP<br>@.5                                      | mAP<br>@.5:.95 | BPR          | Params<br>(M) | GFLOPs       | FPS          | Very Tiny<br>( $< 16^2$ )       | Tiny<br>( $16^2\text{--}32^2$ ) | Small<br>( $32^2\text{--}96^2$ ) | Med/Lrg<br>( $> 96^2$ ) | Total<br>Recall |
| <b>UAVDT Benchmark</b> ( $1280 \times 1280$ )     |                                                 |                |              |               |              |              | (74,375)                        | (206,015)                       | (86,282)                         | (7,325)                 | (373,997)       |
| (1) ESOD Baseline*                                | 0.385                                           | 0.214          | 0.884        | 35.9          | <b>68.2</b>  | <b>117.8</b> | 79.43%                          | 84.21%                          | 94.54%                           | 59.78%                  | 85.17%          |
| (2) Semantic-only                                 | 0.384                                           | <u>0.217</u>   | 0.906        | 35.9          | 75.0         | <u>113.8</u> | 79.13%                          | 83.91%                          | 94.00%                           | 59.95%                  | 84.82%          |
| (3) Spectral-only                                 | 0.387                                           | 0.209          | <u>0.942</u> | 36.0          | 93.7         | 102.3        | 82.43%                          | 88.72%                          | 96.66%                           | <u>66.80%</u>           | 88.87%          |
| (4) (3) + Channel-Pooled                          | <u>0.394</u>                                    | 0.209          | <b>0.963</b> | 35.9          | 99.3         | 97.6         | <b>86.88%</b>                   | <b>92.23%</b>                   | <b>97.74%</b>                    | 66.43%                  | <b>91.93%</b>   |
| (5) Dual-Concat                                   | 0.371                                           | 0.205          | 0.919        | 35.9          | 83.9         | 107.2        | 79.09%                          | 85.85%                          | 95.85%                           | 62.66%                  | 86.35%          |
| (6) Dual-Max                                      | <b>0.395</b>                                    | <b>0.218</b>   | 0.940        | 35.9          | 90.1         | 102.3        | <u>85.69%</u>                   | <u>89.97%</u>                   | <u>97.37%</u>                    | 65.99%                  | <u>90.36%</u>   |
| (7) <b>HESOD (Ours)</b>                           | <u>0.394</u>                                    | 0.215          | 0.940        | <b>26.0</b>   | <u>74.9</u>  | 106.1        | 80.76%                          | 88.98%                          | 97.11%                           | <b>69.16%</b>           | 88.83%          |
| <b>SeaPerson Benchmark</b> ( $2048 \times 2048$ ) |                                                 |                |              |               |              |              | (82,417)                        | (174,816)                       | (42,987)                         | (155)                   | (300,375)       |
| (1) ESOD Baseline*                                | 0.750                                           | 0.320          | 0.947        | 35.8          | <b>202.4</b> | <b>85.7</b>  | 74.03%                          | 86.78%                          | 94.74%                           | 79.35%                  | 84.42%          |
| (2) Semantic-only                                 | 0.769                                           | 0.325          | <u>0.991</u> | 35.8          | 266.8        | 73.5         | 75.49%                          | 91.57%                          | 95.71%                           | 81.94%                  | 87.75%          |
| (3) Spectral-only                                 | 0.770                                           | <u>0.327</u>   | <b>0.992</b> | 35.8          | 267.4        | 71.7         | 75.23%                          | <u>91.69%</u>                   | <b>95.89%</b>                    | <u>86.45%</u>           | 87.77%          |
| (4) (3) + Channel-Pooled                          | 0.767                                           | 0.324          | 0.988        | 35.8          | 263.4        | 72.7         | <b>76.13%</b>                   | 91.36%                          | 95.50%                           | <b>87.10%</b>           | 87.77%          |
| (5) Dual-Concat                                   | 0.772                                           | 0.326          | 0.986        | 35.8          | 281.2        | 69.8         | <u>76.00%</u>                   | 91.50%                          | 95.68%                           | 80.00%                  | 87.84%          |
| (6) Dual-Max                                      | <b>0.778</b>                                    | <b>0.330</b>   | <u>0.991</u> | 35.8          | 255.1        | 73.8         | <u>75.65%</u>                   | <b>92.06%</b>                   | <u>95.85%</u>                    | 81.94%                  | <b>88.10%</b>   |
| (7) <b>HESOD (Ours)</b>                           | <u>0.773</u>                                    | <u>0.327</u>   | <u>0.991</u> | <b>25.9</b>   | <u>208.5</u> | <u>80.6</u>  | 75.71%                          | 91.47%                          | 95.72%                           | 80.65%                  | 88.83%          |

\*Baseline uses BCE only; others add  $\mathcal{L}_{\text{cover}}$  on top. Parenthetical values denote ground-truth counts per bin. One-to-one matching at conf  $\geq 0.001$ , IoU  $\geq 0.5$ . **Bold**: best; underline: second-best.

9.8, and 13.7 pp higher than Faster R-CNN, RetinaNet, and QueryDet respectively at  $5.0\times\text{--}5.6\times$  higher FPS (106.1 vs. 18.8–21.1); on SeaPerson, mAP@.5 exceeds all baselines by 14.6~30.0 pp at  $2.9\times\text{--}5.8\times$  higher FPS, with Very Tiny recall reaching 75.71%.

Against the same-architecture selective baseline ESOD [7]: on UAVDT, mAP@.5 increases from 0.385 to 0.394 (+0.9 pp), BPR from 0.884 to 0.940 (+5.6 pp), total recall by +3.66 pp; parameters are reduced from 35.9M to 26.0M (−27.5%) at 106.1 FPS. On SeaPerson, mAP@.5 increases from 0.750 to 0.773 (+2.3 pp), BPR reaches 0.991 (+4.4 pp), total recall increases by +3.33 pp, successfully recalling more than 10,000 micro maritime targets missed by ESOD; GFLOPs only increase by 3.0%, parameters are compressed by 27.6% at 80.6 FPS. This modest GFLOPs overhead is consistent with the selector routing more target regions downstream; HESOD controls it within 10% via the lightweight head design while maintaining high patch coverage (94.0%/99.1% BPR on UAVDT/SeaPerson).

### 4.3. Ablation Studies and Scale Analysis

We conduct controlled ablation experiments on the ESOD baseline by adding components incrementally, with results in Table 2.

**Dual-Evidence Fusion.** Using only the semantic branch yields 0.906 BPR and 84.82% total recall on UAVDT, limited by semantic feature collapse during downsampling; the spectral-only branch reaches a stronger 0.942 BPR and 88.87% total recall, since gradient features stay robust on micro targets. Comparing fusion strategies, linear concatenation (config. 5) achieves only 0.371 mAP@.5—lower than either single branch—because weak semantic responses introduce destructive interference that suppresses confident spectral activations. In contrast, element-wise maximum (Dual-Max, config. 6) implements an evidence-preserving OR gate, reaching 0.395/0.778 mAP@.5 on UAVDT/SeaPerson—the highest of all seven configurations, including the final compressed HESOD.

**Soft-Coverage Loss.** The ESOD baseline trained with per-pixel BCE achieves only 0.884 BPR and 79.43% Very Tiny recall on UAVDT. With the proposed  $\mathcal{L}_{\text{cover}}$ , the semantic branch’s Very Tiny recall is flat at 79.13%, while the spectral branch’s rises to 82.43%; under Dual-Max fusion it reaches 85.69% (+6.26 pp). The noisy-OR formulation provides strong gradients to low-scoring covering cells when a target is under-covered, encouraging at least one cell to

exceed the routing threshold and reducing unrecoverable misses.

**Channel Pooling.** Reducing the spectral branch to 2 channels (config. 3→4) leaves accuracy essentially unchanged—UAVDT mAP@.5 and BPR both improve (0.387→0.394, 0.942→0.963), lifting total recall by +3.06 pp (88.87%→91.93%), while SeaPerson stays roughly flat (0.770→0.767, 0.992→0.988)—while greatly cutting the memory needed to train on high-resolution inputs like SeaPerson’s  $2048 \times 2048$  frames.

**ISPP Head Efficiency.** ISPP fine-tuning compresses Dual-Max’s peak-accuracy selector, aiming to preserve its accuracy while recovering FPS and cutting parameters. On UAVDT, ISPP cuts parameters by 27.5% and GFLOPs by 16.9% to 74.9, boosting FPS from 102.3 to 106.1, with mAP@.5 down by only 0.001 and BPR unchanged at 0.940. On SeaPerson, ISPP cuts parameters by 27.6% and GFLOPs by 18.2%, lifting FPS from 73.8 to 80.6 with mAP@.5 down by just 0.005 (0.778 → 0.773). The remaining GFLOPs are consistent with the increased number of retained target regions requiring downstream classification and regression.

**Latency Overhead.** FPS is wall-clock inference throughput at batch size 1, averaged over the test set at each dataset’s canonical resolution on an RTX 5090. The modest end-to-end FPS drop versus ESOD (−9.9%/−6.0% on UAVDT/SeaPerson) is therefore consistent with the higher routing recall and increased downstream workload: dual-evidence routing recalls more true targets per image, each of which must still be classified and regressed downstream.

## 5. CONCLUSION

We presented HESOD, uniting semantic objectness with constant-width spectral saliency via monotonic max-fusion to address early semantic collapse on micro targets. An object-level soft-coverage loss ( $\mathcal{L}_{\text{cover}}$ ) optimizes instance coverage under asymmetric pruning risk, while a lightweight ISPP head amortizes downstream overhead without sacrificing selector gains. On UAVDT and SeaPerson, HESOD improves mAP@.5 by 0.9 pp and 2.3 pp (total recall +3.66 pp and +3.33 pp), cuts parameters by >27%, and achieves 106.1/80.6 FPS on an RTX 5090. Dual-evidence routing reaches 0.940/0.991 BPR, substantially reducing routing-stage misses and shifting the remaining bottleneck to downstream regression. These results confirm that spectral structural priors enable accuracy-efficiency co-design for resource-efficient deployment.

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